

Task 3: Lecture Slide Generation Report

1. AI Tools & Models Used

- **Primary Tool:** Gemini 3.1 Pro (Gemini Web interface).
- **Capabilities Utilized:** Multimodal analysis (Video-to-Text), content summarization, and structured data generation.

2. Process Followed

1. **Content Extraction:** I provided the direct YouTube link to Gemini. The model used its multimodal capabilities to watch the orientation lecture, transcribe the key points, and identify the logical structure of the session.
2. **Instruction Framing:** I used a specific prompt requesting a professional, dark-themed presentation with clear section transitions.
3. **Iterative Refinement:** After the first draft, I refined the slide layouts to ensure a balance between visual impact (images/icons) and technical information.
4. **Final Polish:** The output was translated into high-end HTML/CSS layouts to ensure a modern, "consulting-grade" aesthetic.

3. Prompts Used & Effectiveness

- **Initial Prompt (Highly Effective):**

"I have an assignment to generate presentation slides from a lecture video. Here is the link: [YouTube Link]. Please analyze the content and create a 12-slide modern dark-themed presentation. Focus on key learning paths, the registration process, and the value of skill badges."

- **Why it worked:** It provided clear context (assignment), a specific source (URL), a desired length (12 slides), and a specific visual style (dark-themed).

- **Refinement Prompt (Effective):**

"Incorporate more data-driven slides, such as employability metrics and a clear 4-week timeline for the program."

- **Why it worked:** It narrowed the focus on high-value data points that students care about.

- **Failed Attempt (Example of what didn't work):**

"Summarize the video into slides."

- **Why it didn't work:** Too vague. The model produced simple bullet points without a cohesive design or slide-by-slide structure.

4. Prompt Refinement Strategy

The key to refining the prompts was moving from **General Commands** to **Constraint-Based Instructions**. By defining the *audience* (students), the *tone* (inspiring and professional), and the *structure* (timeline, metrics, paths), the LLM was able to generate content that felt like a cohesive presentation rather than a simple summary.

YouTube Video Link: [Video](#)

Slide Link: [Slides](#)